
TECHNICAL REFERENCE

netzspec-api

A fact store for network hardware specifications, where every value carries the document, the table cell and the raw sentence it came from — or is refused.

88,960

LIVE PARTS

107,609

CURRENT FACTS

573

DICTIONARY FIELDS

7,188

SOURCE DOCUMENTS

15,937

OPEN CONFLICTS

52,508

PART RELATIONS

Node 22 · TypeScript ESM · Fastify · PostgreSQL 16 · Python 3.11 adapters
api.netzspec.com · build 7278eb7 · schema at migration 0011

Every figure in this document was read from the production database, the live API, or a pipeline run executed on 5 September 2026.

SECTION 1

What the system is for

A network switch is bought on numbers: switching capacity, forwarding rate, PoE budget, port layout, operating altitude, MTBF. Those numbers are published across vendor datasheets in HTML and PDF, distributor listings and aggregator pages. They disagree with each other, restate imperial units in metric mid-sentence, and routinely place a model number where a port count belongs.

netzspec-api is the store that resolves that into something a machine can query. It holds **89,099 part records** (88,960 live, 139 retired) across 13 vendors and 23 categories, and **107,609 current facts** about them. It is not a scraper with a database attached: the acquisition layer is one input, and the majority of the code exists to decide which number to believe and to make that decision auditable afterwards.

Everything in the design follows from a single constraint: **a value with no traceable source is not data**. A fact that cannot name the document and the table cell it came from is not stored. A parser that cannot read a value returns a reason rather than a guess. Two sources that disagree produce a held conflict, not a winner decided by which process wrote last.

1.1 Why the discipline is not optional

The rules below are not stylistic. Each was written after a specific failure in this codebase, and the cost of each is recorded in the repository:

- A port-layout parser passed 21 hand-written test cases, then read its first real listing — "Catalyst 2960-X 24 GigE PoE" — as **296 ports**, because the leading number in a real Cisco string is the model, and not one test case had a product name in front of the ports.
- A unit converter opened with "no canonical unit, so nothing to check" and returned before reading the magnitude suffix. Every unit-less count silently dropped it: an IPv4 route table of "360K" was stored as **360** — inside the plausible band, so nothing refused it.
- A locale-blind decimal parse read the English "0.075 kg" as **75 kg** under German rules. It was in band. Only replaying the corpus and reading the output found it.

The pattern is consistent: none of these threw an exception, none failed a test, and every one produced a confident, precise, wrong number. The system is built on the assumption that this is the normal failure mode, not the exceptional one.

SECTION 2

Architecture and deployment

Three machines, with a deliberate split between what stores data and what fetches it.

Tier	Runs on	Responsibility
Store & API	Hetzner VPS, Debian	PostgreSQL 16 (localhost-only), Fastify read API under PM2 on :3021, Caddy terminating api.netzspec.com. Holds every part, fact, document and run.
Pipeline	Either machine	TypeScript ESM under tsx: normalisation, the gate, apply, merge, remerge, renormalize, hygiene. Writes only inside a run row.
Acquisition	Operator workstation, Windows	Python 3.11 adapters driving one real Chrome per source through Playwright. Never runs on the server: it needs a real browser profile and the server has no display.

2.1 Why acquisition is not on the server

A headless request is refused by most of these hosts. The adapters therefore drive the operator's installed Chrome with a persistent profile per source, which also keeps each lane's cookies and session isolated. The consequence is that the workstation is a component of the system, not a client of it — and its 8 GB of memory is a real constraint: one lane's Chrome costs roughly 500 MB resident and peaks near 1.2 GB of working set, so the supervisor refuses to start a lane below 1,000 MB free.

2.2 Database access from the workstation

Postgres listens on localhost only. The workstation reaches it through an SSH tunnel on port 5433; nothing is exposed publicly, and the API is the sole network-facing surface. Deploys ship `git archive HEAD`, so an uncommitted file cannot reach the server. The build happens in a staging directory, migrations run, and only then is the tree swapped and PM2 restarted — a failed build or migration leaves the live application untouched.

2.3 Schema history

Migration	Applied (UTC)	What it introduced
0001_init	3 Sep 15:44	parts, vendors, categories, facts, runs
0002_seed	3 Sep 15:44	field dictionary and category profiles
0003_acquisition	3 Sep 15:48	sources, fetch_queue, fetches, source_docs
0004_gaps	3 Sep 15:51	completeness and the gap ledger
0005_watchdog	3 Sep 19:53	watchdog_events, source_throughput view
0006_parts_sku_lookup	4 Sep 04:25	normalised SKU lookup index
0007_image_candidates	4 Sep 05:40	image candidates and variants
0008_conflict_raws	4 Sep 07:58	kept_raw / rejected_raw on conflicts
0009_parts_retired	5 Sep 05:52	retirement with successor pointer
0010_parts_case_unique	5 Sep 07:16	case-insensitive part identity per vendor
0011_sources_proxy	5 Sep 09:27	per-source proxy and metered proxy_bytes

All eleven migrations are applied in production. Migrations are forward-only and must be additive: a rollback restores code, never schema.

SECTION 3

The data model

Twenty-six tables. The row counts below are live production values, and the shape of the distribution is itself informative — `fact_evidence` and `doc_parts` outnumber the facts themselves, because provenance is stored per occurrence.

Table	Rows	What it holds
parts	89,099	One row per SKU per vendor. Identity is case-insensitive within a vendor.
facts	132,535	Append-only. 107,609 current, 24,926 superseded.
fact_evidence	139,151	Each document occurrence supporting a fact.

Table	Rows	What it holds
doc_parts	126,375	Which parts a document mentions.
completeness	89,099	Required / present per part, one row per part.
relations	52,508	compatible 35,726 · successor 16,782.
conflicts	44,252	15,937 currently open.
lifecycle	18,021	eol_announced 17,749 · active 272.
fetch_queue	17,224	queued 12,315 · done 4,859 · failed 49 · leased 1.
image_variants	8,116	Generated WebP renditions.
source_docs	7,188	Fetches documents by content hash.
category_profiles	5,519	Which fields are required per category.
fetches	4,609	One row per network fetch, with byte counts.
part_source_checks	3,854	"This source was asked about this part."
images	2,751	Linked product photography.
source_fields	1,046	Which sources publish which fields.
part_aliases	586	Spare, case and variant SKU forms.
field_dictionary	573	Every legal field key, with type and unit.
sources	20	Acquisition lanes.
watchdog_events	29	Every automated pause, back-off and resume.
runs	80	Every write to the store, ever.
categories	23	Product taxonomy.
vendors	13	Manufacturers.
schema_migrations	11	Applied migrations.
api_keys	3	Scoped read keys.

3.1 The catalogue

Product class	Parts	Vendor	Parts	Hardware
hardware	63,276	cisco	86,944	61,270
license	16,664	hpe	478	478
software	7,745	aruba	358	358
unknown	770	arista	344	344
service	505	juniper	168	168
		dell-emc	151	151
		9 others	622	614

Cisco is 97.6% of the catalogue by design — it was the first vendor taken to depth. "license" and "software" are separated from hardware because a licence that reports the port count of the chassis it licenses is a specific, recurring extraction fault.

3.2 Categories

Category	Parts	Category	Parts	Category	Parts
security	13,444	wireless	6,273	video	3,534
servers / UCS	12,854	cloud mgmt	5,685	collab endpoints	3,222
switches	10,792	unified comms	5,447	optical	2,980
routers	8,345	conferencing	3,749	contact centre	2,204

SECTION 4

Anatomy of one fact

This is a live record from the API: the maximum operating altitude of a Catalyst 9200L-24P-4G. Nine of its thirteen fields exist for one purpose — so that a human can disagree with it later and find out who was right.

```
GET /v1/parts/cisco/C9200L-24P-4G -> facts[0]
```

```
{
  "key"          : "altitude_max",
  "label_en"     : "Max operating altitude",
  "label_de"     : "Max. Betriebshoehe",
  "type"         : "n",
  "value"        : 3000,
  "unit"         : "m",
  "raw"          : "10,000 ft. (3000 meters), up to 45 C",
  "state"        : "verified",
  "tier"         : 2,
  "method"       : "html_table",
  "inherited"    : true,
  "inherited_from" : "catalyst-9200-series-switches",
  "source": {
    "doc_id"      : "de692de8d2c8f641",
    "url"         : "cisco.com/.../nb-06-cat9200-ser-data-sheet.html",
    "locator"     : "t18:r15:c1",
    "extracted_at" : "2026-09-02"
  },
  "evidence_count" : 1
}
```

Field	Why it exists
raw	The sentence exactly as printed. The parser read feet, found the metric restatement in brackets and kept 3000 m — and the comma in "10,000" did not become a decimal separator, which is a bug this project has paid for in German-locale text. Keeping the raw string is what makes a re-parse possible without re-fetching: 12,762 facts have already been re-normalised in place from stored raws.
locator	Table 18, row 15, cell 1 of that document. 99,739 of 107,609 current facts (92.7%) carry one. A fact that cannot name its cell cannot be argued with.
doc_id	The content hash of the fetched document. 92,707 facts (86.2%) carry one; the remainder are tier-0 operator seed data, which has no document by definition.
inherited / inherited_from	This value came from the series datasheet, not the SKU's own page, and says so. 36,292 facts are inherited. A value is never quietly inherited into a SKU the document does not list.
tier	The authority of the source. A distributor listing (tier 4) cannot overwrite a vendor datasheet (tier 2), whichever arrived later.
state	verified, corroborated, conflict, unverified, gap_unattempted or not_applicable. It never becomes "deleted".

Field	Why it exists
norm_v	The normaliser version that produced the value, so a parser fix can find every fact it invalidates. Currently 1.0.0 (56,503), 1.5.1 (12,762), 1.5.0 (7,487), 1.4.0 (1,292).
run_id	The run that wrote it. Every fact is attributable to one command invocation.

Facts are never overwritten

A newer reading *supersedes* the old row, which remains in place stamped with the run that retired it. 24,926 rows are currently superseded — 18.8% of all fact rows ever written. That history is what makes it possible to ask not just what a specification is, but what it used to be and why it changed.

SECTION 5

The pipeline

Seven stages. The figures in 5.4 come from a real dry run over the Meraki corpus executed on 5 September; nothing was written, because the apply path refuses to commit unless the gate passes first.

5.1 Fetch

A task is leased from `fetch_queue` with `FOR UPDATE SKIP LOCKED`, so several lanes can work the queue without collision. Politeness is per source and enforced per host (2–12 seconds). `robots.txt` is parsed and respected — one source, CDW, is disabled entirely because it disallows the path the adapter would need. The same URL is never fetched twice in a day unless the task explicitly forces it.

5.2 Cache

Every page is written to disk under the SHA-1 of its URL: **14,836 HTML documents and 669 PDFs** on the workstation, mirrored on the server. A challenge or interstitial page is *never* cached, and a cached one is deleted on sight — a test suite once scored 24 vacuous passes against a cached Cloudflare challenge, which is why that rule is enforced at the cache layer rather than the adapter.

5.3 Extract and normalise

A Python adapter per source reads tables into label/value pairs with a cell locator, and does no cleaning whatsoever — a Python adapter that tidies a value is a second normaliser that will drift from the first. Normalisation is TypeScript: it maps the label to a dictionary key, parses the raw string to a typed value, converts to the field's canonical unit and checks the result against the field's plausibility band. If it cannot, it returns a named reason and stores nothing.

5.4 What one real run produced

Stage	Count	Meaning
Pages read	96	Meraki corpus, from cache, no network
Facts accepted	1,458	typed, united, within band, with provenance
Facts refused	227	each with a named reason (below)
Labels unmapped	216	no dictionary entry yet; recorded, not guessed
Sentinel values	65	"N/A", "-", "varies" — recognised, not stored

Stage	Count	Meaning
SKUs resolved	130	all by exact match; 0 ambiguous
Image candidates	122	2 refused on sight as placeholders
Gate precision	1.00	sampled against a hand-checked suite
Gate recall	1.00	no known fact was missed in the sample

The 227 refusals, by reason

Reason	n	What it means
tdp:PARSE_FAIL	33	thermal design power in a form the parser will not risk
tdp:UNIT_MISSING	32	a number with no unit, where the field requires one
dimensions:STRUCT_UNPARSED	32	a structured field with no dedicated parser yet
tdp:UNIT_UNKNOWN	31	a unit not in the dictionary for that field
cooling:ENUM_VIOLATION	17	a value outside the field's declared enumeration
poe_budget:PARSE_FAIL	16	unparseable PoE budget
mgig_rj45_ports:PARSE_FAIL	16	multi-gigabit port count not safely readable
sfp_plus_ports:PARSE_FAIL	12	SFP+ port count not safely readable
mtbf:UNIT_MISSING	6	hours implied but not stated
poe_standard:ENUM_VIOLATION	3	PoE standard outside the enumeration

Refusals are the interesting half of the output. A recorded gap is recoverable at any time; a plausible wrong number propagates into every downstream consumer and is found only by accident.

A required field nothing can fill is a permanent gap, not a recorded one

The ports field was marked required for switches while its parser branch ended in STRUCT_UNPARSED — a deliberate refusal to guess. The reasoning was right and the consequence was not: every switch reported a gap on the one specification a switch is bought for, no extraction could ever close it, and it was quietly the largest single rejection class in the pipeline. Refusing to guess is correct; leaving a required field unfillable is a decision to fail forever. 3,520 ports facts now exist.

SECTION 6

The gate

No run writes facts without a passing gate, and the gate runs inside the same transaction as the write.

The gate samples the run's output against hand-checked suites and measures precision (is what we produced correct?) and recall (did we miss what the page plainly states?). Below threshold, the entire run rolls back — not the offending fact, the run. Of 80 runs in the store, **14 failed or aborted** and left nothing behind, which is the mechanism working rather than a fault.

Run kind	Succeeded	Failed	Other	Purpose
apply-acquired	29	8	1 running	scraped pages to facts
sync-dictionary	8	—	—	field dictionary into the database

Run kind	Succeeded	Failed	Other	Purpose
apply-remerge	7	—	—	re-evaluate open conflicts
apply-specs	6	4	1 aborted	Cisco datasheet extraction
hygiene-*	5	—	—	duplicates, foreign PIDs, variants
images	3	1	—	product photography
migrate-atlas	2	1	—	one-time load from MongoDB
apply-renormalize	2	—	—	replay the normaliser over stored raws
apply-enumeration	1	—	—	enumerated PIDs become parts
reclassify	1	—	—	re-run the product-class rules

6.1 Proof by sabotage

Every gate ships with a deliberately broken input that must be rejected *for the stated reason*. A rejection for the wrong reason counts as a miss, because an error message that sends an engineer to fix the wrong thing is the same as no check at all. To confirm a check is alive, it is disabled and the suite must go red; if the suite stays green, the check was already dead.

This is not theoretical. A placeholder-detection rule was written, reviewed, committed and reported "all checks passed" — while matching nothing at all, because it had been written through a shell heredoc that turned its word boundaries into literal backspace characters. It compiled. It ran. Every tool that displayed it showed it as correct. It was found only by dumping raw bytes.

Suite	Cases	Covers
tests/scrapper/test_watchdog.py	248	watchdog rules, sentinel, supervisor script AST
tests/scrapper/test_worker_units.py	128	queue, outcomes, proxy metering, budget
tests/db/hygiene.test.ts	100	catalogue hygiene, 42 sabotage cases
tests/db/remerge.test.ts	73	conflict re-evaluation under current rules
tests/db/apply-enumeration.test.ts	48	PID enumeration, 13 sabotage cases
tests/scrapper/test_worker_browser.py	22	browser launch arguments, route filter

Plus the pure TypeScript suites and a control-character scan of the source tree, which exists specifically to catch the heredoc failure described above.

SECTION 7

The merge layer and conflicts

Two sources describing the same field of the same part is the normal case, not the exception. Resolving it by write order would make the database a function of scheduling.

7.1 Source tiers

Tier	Kind	Sources	Facts	Authority
0	operator	hexcat	32,484	Protected. Never overwritten automatically.

Tier	Kind	Sources	Facts	Authority
1	vendor	cisco-datasheet-pdf, fortinet, hpe-quickspecs	5,782	Vendor PDF and QuickSpecs.
2	vendor	cisco-datasheets, meraki, arista, juniper, dell, extreme, mikrotik, nvidia, ubiquiti, cisco-eol, cisco-tmg	69,322	Vendor HTML datasheets. The bulk of the store.
3	aggregator	icecat-open, itprice, router-switch	21	Used for discovery and cross-checking, rarely for values.
4	distributor	cdw, provantage	—	Identifiers, UPC/GTIN and images; not specifications.

7.2 Open conflicts

15,937 conflicts are open right now, out of 44,252 ever logged. They are held, not resolved, and the API can serve them per part.

Reason	Count	Reading
higher tier rejected ($2 > 1$) and sources disagree	8,211	A tier-2 datasheet contradicts a tier-1 PDF. Tier alone does not settle it.
same-tier sources disagree; field held	5,972	Two equally authoritative documents disagree. No rule can choose; a human must.
operator-reviewed value is protected (tier 0)	733	A human already decided. Automation may not override it.
SAME_DOC_REEXTRACTION (norm_v drift)	~600	The same document re-read under a newer normaliser produced a different value — a parser change, not a source disagreement.
migrated: held in Atlas without a logged conflict	55	Inherited from the pre-PostgreSQL store.

A resolution class is believed only after its rows are read against the stored pair

A remerge pass once reported 4,164 "exact agreements". Every one had kept != rejected: 2,494 were comparing two null raw strings that re-normalised to the same nothing, and an altitude of 4998.72 against 3000 was being called exact. The report's own per-class samples were printed under the wrong heading and it passed review. Before any bulk resolution is trusted, twenty random rows of the class are read by hand against the stored kept/rejected pair.

7.3 Normaliser versioning

Each fact records the normaliser version that produced it, so a parser fix can locate exactly the facts it invalidates and replay them from stored raw strings — without re-fetching a single page. 12,762 facts currently carry 1.5.1, 7,487 carry 1.5.0, 1,292 carry 1.4.0, and 56,503 still carry the original 1.0.0. Replay is itself gated: a renormalize run that would change more than a quarter of the facts it touches refuses to commit without a written reason.

SECTION 8

Sources and the field dictionary

Source	T	Kind	Host	Delay	Path	On
hexcat	0	operator	local	—	direct	no
cisco-datasheet-pdf	1	vendor	www.cisco.com	2.0 s	direct	no

Source	T	Kind	Host	Delay	Path	On
fortinet	1	vendor	www.fortinet.com	3.0 s	direct	no
hpe-quickspecs	1	vendor	www.hpe.com	3.0 s	direct	no
arista	2	vendor	www.arista.com	3.0 s	direct	no
cisco-datasheets	2	vendor	www.cisco.com	2.0 s	direct	no
cisco-eol	2	vendor	www.cisco.com	2.0 s	direct	no
cisco-tmg	2	vendor	tmgmatrix.cisco.com	2.0 s	direct	no
dell	2	vendor	www.dell.com	3.0 s	direct	no
extreme	2	vendor	www.extremenetworks.com	3.0 s	direct	no
juniper	2	vendor	www.juniper.net	3.0 s	direct	no
meraki	2	vendor	documentation.meraki.com	2.0 s	direct	no
mikrotik	2	vendor	mikrotik.com	2.0 s	direct	no
nvidia	2	vendor	docs.nvidia.com	3.0 s	direct	no
ubiquiti	2	vendor	techspecs.ui.com	2.0 s	direct	no
icecat-open	3	aggregator	live.icecat.biz	1.0 s	direct	no
itprice	3	aggregator	itprice.com	3.0 s	residential	no
router-switch	3	aggregator	www.router-switch.com	12.0 s	residential	no
cdw	4	distributor	www.cdw.com	3.0 s	direct	no
provanage	4	distributor	www.provanage.com	3.0 s	direct	no

All twenty sources are currently disabled — see section 9. CDW is disabled permanently: its robots.txt disallows the search path the adapter would need.

8.1 The field dictionary

573 field keys. A fact cannot be written with a key that is not in this table — a foreign key enforces it, so a typo in an adapter cannot invent a field. Each entry carries a type, an optional canonical unit, English and German labels, and a plausibility band.

Type	Keys	Meaning	Dictionary	Count
s	316	free string	Total keys	573
n	181	number with a canonical unit	With a canonical unit	225
ls	27	list of strings	Generated from evidence	440
e	16	enumeration	Hand-authored	133
b	16	boolean	Categories with profiles	23
nr	10	numeric range (min/max)	Profile rows	5,519
struct	7	structured value		

Most-populated fields

Field	Facts	Field	Facts	Field	Facts
certifications	5,375	series	3,286	form_factor	2,943
temp_operating	5,085	supported_protocols	3,274	humidity_storage	2,704
temp_storage	3,994	humidity_operating	3,153	emc_emissions	2,552
vendor	3,876	data_rate	3,016	power_max	2,529
standard	3,846	ieee_standards	3,015	altitude_max	2,513
ports	3,520	mgmt_class	2,460	emc_immunity	2,437

8.2 Extraction methods

Method	Facts	What it is
html_table	43,068	a two-cell row of a vendor specification table
hexcat_seed	32,484	operator-reviewed seed data, tier 0
description_mining	20,567	prose in a product description, parsed conservatively
pdf_table	2,085	a table extracted from a datasheet PDF
product_name_mining	1,128	facts derivable from the product name itself
retracted:family:mismatch	1,797	withdrawn: inherited into a SKU the document did not list
retracted:class:license	1,645	withdrawn: a licence reporting its chassis's specifications
retracted:component:SFP	1,233	withdrawn: a transceiver reporting its host switch's specifications

The three retraction classes are facts the system withdrew from itself after the rule that produced them was found to be wrong. They remain in the table, labelled, rather than being deleted.

SECTION 9

Acquisition, and why it is currently stopped

All twenty sources are disabled. This section records what was tested on 5 September 2026 and what the results establish.

9.1 The residential proxy

Two aggregator lanes were refusing this workstation's IP. A metered residential proxy was built for them: `sources.proxy` and `proxy_country` decide per lane, credentials are parsed in exactly one function and rendered only through a redactor that cannot return a password for any input, `images/media/fonts` and `analytics` hosts are aborted before they cost anything, and every response is metered into `fetches.proxy_bytes` against a per-source daily budget.

The machinery works. On a single router-switch product page the route filter aborted **148** image, font and analytics requests. The gateway exits from a genuine US residential address (verified: `proxy:false`, `hosting:false`). Nine metered fetches recorded 1,770,274 bytes — 1.69 MB of a 5,120 MB plan.

9.2 What the proof runs found

Lane	Network path	Result, 3 tasks each	Verdict
itprice	residential, US exit	3 of 3 challenged (twice, incl. re-seeded profile)	blocked
router-switch	residential, US exit	1 answered, 2 challenged	blocked
provanage	direct — workstation IP, no proxy	3 of 3 challenged	blocked

All three sites return an **interactive Cloudflare Turnstile** — "Performing security verification / Verify you are human", with a checkbox a person must tick. It is not the automatic JavaScript challenge that clears on its own; the worker waits 25 seconds and re-reads the DOM ten times, and it never clears.

The finding that matters

provanage was challenged on the workstation's own IP with no proxy in the path at all. That rules out IP reputation, proxy configuration, pool size and exit country as the cause — a direct lane never touches the proxy provider. The sites have raised their posture against automated browsers generally. Getting past an interactive challenge means defeating the bot detection itself, which this project will not do. The lanes are paused with the reason recorded, and the realistic options are a data feed or written permission from the sites, or replacing distributor lanes with official vendor sources — which are tier 2 and outrank distributors in every merge anyway.

9.3 What still works without the network

The cache holds 14,836 HTML documents and 669 PDFs, and 7,188 documents are registered in the store. Extraction, normalisation, the gate, apply, merge, remerge, renormalize, hygiene, completeness and the entire API run against that corpus with no network access at all. The Meraki run in section 5.4 was executed this way. Acquisition being stopped constrains growth; it does not stop the system.

SECTION 10

Monitoring

Two monitors, deliberately separated by the question they answer.

Monitor	Question	Acts?
watchdog.py	Is the data healthy? Yield per source, adapter drift, landing measured at the database, apply failures, stale runs, duplicate fetches, unmapped labels, proxy spend.	With --act: pause, back off, resume.
sentinel.py	Is anything physically alive? Supervisor, SSH tunnel, one worker per enabled source, one Chrome per worker, heartbeat age, RAM.	With --heal: restart lanes, kill wedged ones.

10.1 Rules learned from monitor failures

- **A monitor that cannot tell its own limit from a fault sends someone to fix nothing.** A link checker once fired 23 requests back to back, tripped the host's rate limiting, and reported 23 broken pages on a site where every page loaded. There is now a third state — unverified — distinct from pass and fail.
- **A monitor's own failure must be loud.** The sentinel died silently on a 60-second PowerShell timeout and every lane sat idle for two hours. Its crash guard now covers the whole cycle, including writing the report, and a cycle that could not write says so at the top of the next one.

- **One file per writer.** Both monitors wrote and deleted the same ALERT.md from their own alarms, so a clean cycle of either erased the other's live ones and whichever ran last decided what the night looked like. Each now owns exactly one file and the merged summary is rebuilt from disk.
- **Read stats, not headlines.** A hygiene run printed "COMMITTED ... merge 8" with "failed: 8" in its own statistics block.

10.2 A gap this document should record

The watchdog's block alarm fires at five blocks in the window. In the 5 September runs itprice reached six and was reported as BLOCKED 6x; provantage reached three and produced *no block line at all*. A source can therefore be completely dead and stay quiet. Furthermore the block was detected by the HTTP 403 status rather than by a challenge fingerprint — and the current fingerprint pattern does not match the wording Cloudflare now uses, so a challenge served with HTTP 200 is still classified by the adapter as "this part is not listed". Both are open work.

10.3 Automated actions taken to date

Event kind	n	Meaning
stall	16	a lane stopped moving while work was runnable
zero_yield	6	pages fetched, no facts landed
resumed	3	a back-off expired and the lane was re-enabled
source_backoff	2	blocks crossed the threshold; queue deferred 60 min
junk_deleted	1	queue keys that were not part numbers
source_paused	1	adapter drift; lane disabled pending a human

Every automated change to a source is an event row carrying the numbers that justified it. 29 in total.

SECTION 11

The read API

Fastify, key-scoped, read-only. 29 routes under /v1.

Route	Returns
/health	liveness, database reachability, git SHA of the running build, part count
/parts	filtered, paginated part list; cursor-based
/parts/{vendor}/{sku}	one part: facts, provenance, images, completeness, lifecycle, relations
/parts/{vendor}/{sku}/facts	the fact list alone
/parts/{vendor}/{sku}/history	every superseded reading, with the run that changed it
/parts/{vendor}/{sku}/conflicts	open disagreements, with both sides and their evidence
/parts/{vendor}/{sku}/gaps	required fields not yet present
/parts/{vendor}/{sku}/similar	comparable parts
/parts/{vendor}/{sku}/successors	replacement parts after end-of-life
/search	text search across the catalogue
/families, /families/{vendor}/{family}	product families

Route	Returns
/compare	several parts side by side
/facets	aggregate counts for filter construction
/fields	the field dictionary
/categories, /vendors	taxonomy
/lifecycle	end-of-life feed
/changes	what changed, by run
/stats, /stats/gaps	coverage and completeness
/runs, /runs/{id}	the write history of the store
/docs/{doc_id}	a source document's metadata
/export	bulk export
/tools, /tools/{id}, /tools/{id}/run	126 named finders over the catalogue

11.1 Strict query parsing

An undeclared query parameter is a **400** naming the offending key and the accepted set, derived from each route's own schema. This is a deliberate reversal: `?sku=` had been silently ignored for weeks, so every caller using it received unfiltered results and had no way to discover the mistake. A parameter that is silently dropped is worse than one that errors.

11.2 Known API gaps

- Every parts read path currently serves retired rows. `/health` reports 89,099 parts where 88,960 are live; the 139 retired rows are not filtered yet. A retired SKU should 404 with its successor in the body.
- `has=facts` and `filter=` do not yet apply the succeeded-run rule that the other read paths use, and neither do `facets`, `stats`, `gaps`, `compare` or `changes`.
- `/health` sits outside the strict-query hook.
- A duplicate range parameter is silently ANDed rather than rejected.
- Not yet built: a per-part specification sheet grouped by profile section in German and English, a family comparison matrix, and lookup by any identifier (UPC/GTIN, distributor SKU, case variant, hardware twin).

SECTION 12

Coverage, and what is honestly missing

Completeness is measured per part as required fields present over required fields defined by that part's category profile. The catalogue-wide average is **21.6%**.

Completeness band	Parts	Share	Reading
0%	50,665	56.9%	no facts at all — overwhelmingly un-crawled Cisco PIDs
1–19%	7,932	8.9%	identity and a little else
20–39%	8,375	9.4%	partial datasheet coverage

Completeness band	Parts	Share	Reading
40–59%	2,302	2.6%	most of a datasheet read
60–79%	13,491	15.1%	good coverage
80–100%	6,334	7.1%	effectively complete

The distribution is bimodal, and that is the honest headline: where a datasheet has been read, coverage is good — 19,825 parts sit above 60%. The 50,665 parts at zero are almost entirely enumerated Cisco PIDs that exist as catalogue entries and have never had a document fetched for them. Closing that gap is an acquisition problem, which section 9 explains is currently blocked for distributor and aggregator lanes but *not* for official vendor sources.

12.1 Data faults recorded but not yet fixed

Filed, reproducible, and deliberately not hidden:

- Roughly five c9550-* rows have a temperature sentence mapped to altitude_max — a label-mapping fault.
- sfp_ports is typed as a string on some rows and holds "-".
- 3,632 ports and upLink_ports values are strings under a struct type.
- 22 parts carry a family naming a different manufacturer.
- manageable reads "n/a" as false rather than as unknown.
- Source locale is recorded nowhere; hexcat seed data is assumed German by rule.
- 4,282 conflicts are drift artefacts where the current fact is the value a corrected rule rejected — write-order pollution from before the rule was fixed. They need a dedicated repair class that restores the recorded kept value and re-evaluates.

12.2 Ordered backlog

#	Item	Why it is where it is
1	Challenge fingerprinting and a per-source block-rate alarm	A dead source is currently silent below five blocks.
2	Filter retired parts on every read path	The API serves rows it knows are retired.
3	Drift repair class for 4,282 conflicts	Largest single block of resolvable conflicts.
4	List-field renormalize under normaliser 1.5.2	About 2,254 list conflicts are waiting on it.
5	Specification sheet and lookup endpoints	The two most requested missing API surfaces.
6	Official Cisco lanes running permanently	Datasheets, EoL bulletins and the TMG matrix are tier 2 and not blocked.
7	Adapter recall audit	Measure each adapter against an independent inventory of what its pages hold.

APPENDIX

Glossary and operating notes

Term	Definition
fact	One field of one part, with a value, unit, raw source string, state, tier, method, document, locator and the run that wrote it.
supersede	Replace a fact by writing a new row and stamping the old one, never by updating in place.
tier	Source authority, 0 (operator) to 4 (distributor). Lower is stronger.

Term	Definition
state	verified, corroborated, conflict, unverified, gap_unattempted, not_applicable.
run	One command invocation that may write. Carries inputs with hashes, a gate result, statistics and a git SHA.
gate	The precision/recall check a run must pass before its writes commit.
landing	Yield measured at the database rather than at the adapter: how many facts actually reached a part.
locator	Position within a document, e.g. t18:r15:c1 for table 18, row 15, cell 1.
norm_v	The normaliser version that produced a value.
sentinel value	A non-value such as "N/A", "-" or "varies", recognised and not stored.
profile	Per category, which fields are required. Scores completeness; must never filter what is served.
drift	An adapter reading measurably less per page than it did over the previous week.
canary	A known-good URL fetched every monitoring cycle to detect a block in one cycle rather than by inference. Specified, not yet built.

Operating invariants

- Never write to the database outside a run.
- Never overwrite a fact — supersede it.
- Never resolve a disagreement by write order.
- Never inherit a family value into a SKU the document does not list.
- Never guess a value; return a reason and quarantine.
- Never add a field key by hand — add it to the dictionary, where a foreign key enforces it.
- A check that has never failed is not a check.
- Run over the real corpus and read the output before believing a green suite.

A closing note on the numbers in this document

Every figure here was read from the production database, the live API or a pipeline run on 5 September 2026. Where two numbers disagree — the 89,099 parts reported by /health against the 88,960 live parts on the cover — both are printed and the discrepancy is explained, because a document that rounds away its own inconsistencies is not a reference. This one is generated from queries, so it can be regenerated whenever the numbers move.